



Customer Shopping Behavior Analysis

*Uncovering insights from **3,900 transactions** across product categories — analyzing spending patterns, customer segments, product preferences, and subscription behavior to guide strategic decisions.*

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Key Feature Groups

- *Demographics: Age, Gender, Location, Subscription Status*
- *Purchase Details: Item, Category, Amount, Season, Size, Color*
- *Behavior: Discounts, Promo Codes, Frequency, Review Rating, Shipping*

Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean & Standardize

Imputed missing Review Ratings with category median; renamed columns to snake_case.

03

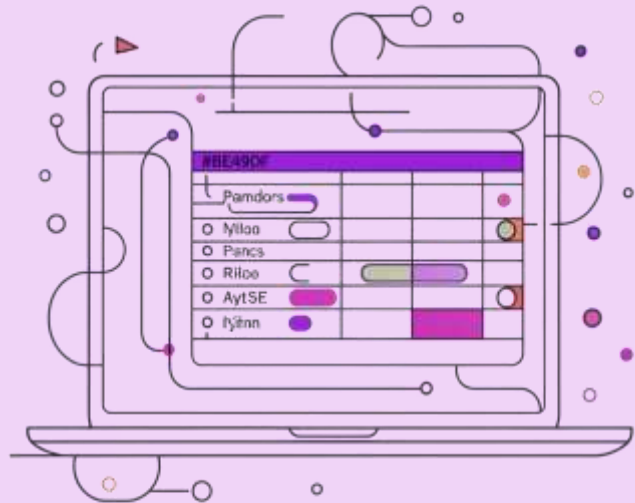
Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

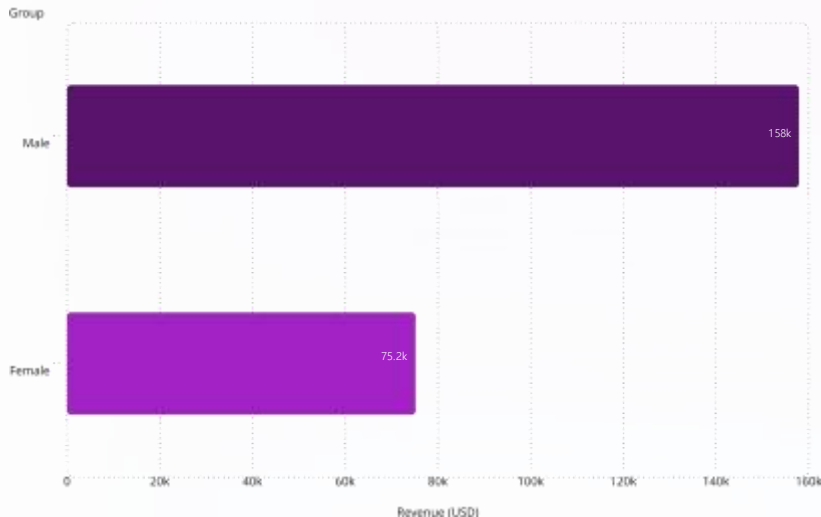
04

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for structured SQL analysis.



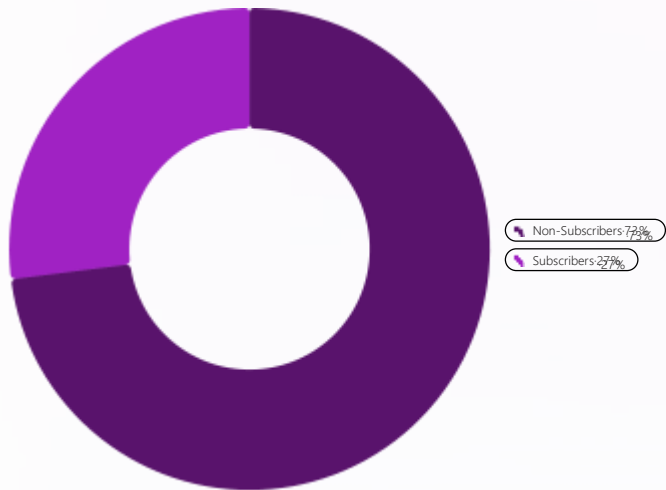
Revenue & Spending Insights



Key Findings

- *Male customers generated \$157,890 vs. Female at \$75,191 in total revenue.*
- *Express shipping users averaged \$60.48 vs. \$58.46 for Standard — higher spend with faster delivery.*
- *839 discount users still spent above the average purchase amount, signaling strong value perception.*

Subscription vs. Non-Subscriber Behavior



Spend Comparison

Status	Customers	Avg Spend	Total Revenue
Subscribers	1,053	\$59.49	\$62,645
Non-Subscribers	2,847	\$59.87	\$170,436

Average spend is nearly identical — but **73% of customers are non-subscribers**, representing a major conversion opportunity.

Top Products & Discount Dependency

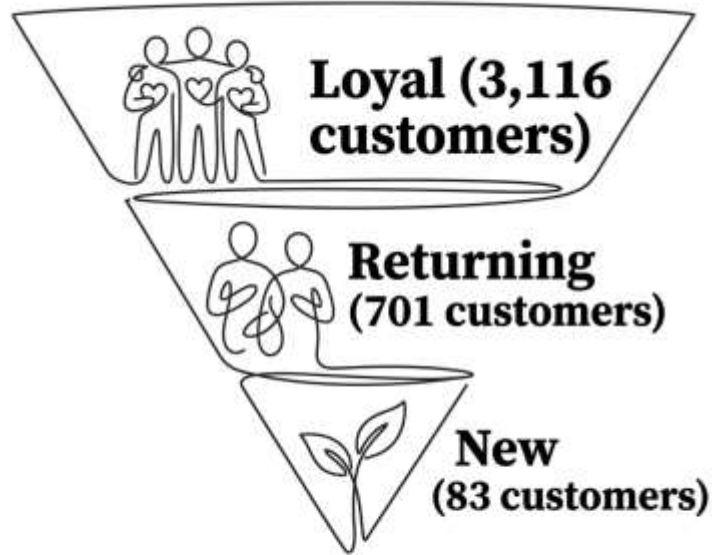
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation & Loyalty



**showing a large loyal base
with few new customers**

Segment Breakdown

Loyal — 3,116

Dominant segment with >5 purchases. 958 are also subscribers.

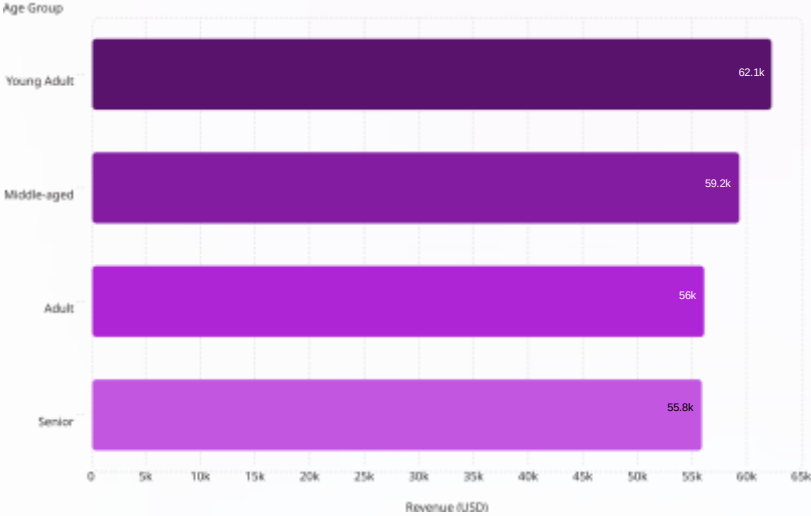
Returning — 701

Engaged but not yet loyal — prime targets for loyalty programs.

New — 83

Small new cohort; focus on onboarding and first-purchase experience.

Revenue by Age Group & Top Products per Category



Top Products by Category

Category	#1	#2	#3
Accessories	Jewelry	Sunglasses	Belt
Clothing	Blouse	Pants	Shirt
Footwear	Sandals	Shoes	Sneakers
Outerwear	Jacket	Coat	—



Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Interactive filters by Subscription Status, Gender, Category, and Shipping Type. Visualizations include revenue & sales by category and age group, plus subscription status breakdown.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on high-discount products.



Targeted Marketing

Focus on high-revenue age groups, top-rated products, and express-shipping users.